



Fair travel distances in tournament schedules: A cooperative game theory approach

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ABSTRACT

The most studied problem in sports scheduling, so-called traveling tournament problem (TTP), aims at finding schedules minimizing the total distance traveled by the teams. While minimizing all the traveling between games is efficient from the overall perspective, it overlooks the distribution of the travel among the teams. Consequently, some teams may end up better than others with respect to their individual goals, an imbalance which may affect teams' often-limited resources or preparedness for the games. This article adopts a cooperative game theory framework to obtain tournament schedules where the distances traveled by the teams are allocated according to fairness criteria. The approach consists of three steps. First, the scheduling problem is reformulated as a transferable utility game. Second, by means of well-established allocation methods, an ideal distance distribution among the teams is determined. Third, we introduce fairness measures to produce a schedule which approximately resembles the ideal distribution. We also discuss the case of not pursuing fairness, but rather a compromise between fairness and minimum total distance. We illustrate the approach by a numerical example in one of the classic TTP data instances.

1. Introduction

Due to its importance in practice and the methodological challenges it involves, scheduling sport tournaments has received a lot of attention from researchers and practitioners. While random approaches or basic templates can capture the most essential requirements to schedule a tournament, more sophisticated approaches have attempted to incorporate more criteria. These criteria often derive from logical and operational conditions (such as the number of matches between each pair of opponents and time-slot availability), and from league-specific requirements. Then, these criteria are put together into an optimization problem, where some of the conditions are modeled as constraints and the others as part of the objective function (see [Van Bulck et al. \(2020\)](#) for a recent survey documenting typical constraints and objective functions in the literature). Whether it is a minimization or a maximization problem, the performance of alternative solutions is usually measured by an overall goal aggregating all teams. Examples of optimization criteria include the minimization of total travel distances, the maximization of total revenues from attendance and TV, and the minimization of total penalty costs for not satisfying some constraints. In consequence, some teams may end up better than others with respect to their single goals. It

is therefore interesting to address how the benefits of the overall best solution could be fairly split among the different teams. Cooperative game theory offers a framework to deal with that question. This article focuses on the intersection of such cooperative game theory framework and fairness in sports scheduling, with particular emphasis on fairness with respect to travel distances in the traveling tournament problem (TTP).

Introduced in the seminal work by [Easton et al. \(2001\)](#), the TTP is the most studied problem in the sports scheduling literature. The objective of the TTP is to find a schedule that minimizes the total traveled distances incurred by the teams, while satisfying other essential constraints of the tournament. Some real-world applications of this problem are reported in [Bonomo et al. \(2012\)](#) and [Durán et al. \(2019\)](#). Despite these and other numerous works on methodological approaches to solve the TTP, the discussion of fairness issues around this problem has not received much attention.

In the more general context of sports scheduling, fairness is important because it may affect the effort put by a team to win a match, as shown by [Krumer and Lechner \(2017\)](#) and [Chater et al. \(2021\)](#). [Van Bulck and Goossens \(2020\)](#), [Durán et al. \(2021, 2022\)](#) have produced recent efforts to create fair schedules, focusing on rather specific tournament designs

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where the fairness criteria are *ad hoc* to such designs. However, more general and theoretical approaches to define fairness in sports scheduling and particularly in the context of the TTP are still lacking in the literature.

In this article, we address the problem of fairness in scheduling sports tournaments, where fairness refers to the travel distances of the teams. Our contribution is two-fold. First, we propose a cooperative game theory approach to obtain the ideal traveling distances for every team, showing that the optimal solution of the TTP is not necessarily the fairest solution. Second, as finding a schedule which renders exactly those ideal distances might not be viable, we use some fairness metrics to find schedules that prescribe distances close to the ideal. Based on the Pareto efficiency principle, we then suggest a way to find a compromise solution between the schedule minimizing the total distance and the schedule maximizing fairness.

To the best of our knowledge, our article is the first one adopting a cooperative game theory framework to address fairness in sports scheduling. This is a well-established framework founded on general principles of fairness which have been studied in a broad range of applications (see e.g. Curiel (2013)). Also, by focusing on the flagship problem of the area, namely the TTP, our article brings the fairness discussion further to a common venue of research where the main focus has instead been on algorithmic approaches to solve complex optimization problems. New algorithmic development is by all means valuable, but it is outside the scope of our paper (interested readers are referred to Van Bulck and Goossens (2023) for a recent overview on methodologies to solve challenging sports timetabling problems). In contrast, there is not much work on cooperative game theory in sports tournaments, where most of the game theory work has focused on its non-cooperative branch. This is perhaps because a sport tournament is essentially a competitive situation. However, we may argue that the top teams of top leagues would not be what they are without *smaller* teams taking part in the same tournaments. The success of the tournaments is subject to the participation of all teams, which motivates the introduction of a cooperative game theory perspective. In this regard, recent contributions include for example Bergantiños and Moreno-Ternero (2020, 2021, 2023), which focus on sharing the revenues from broadcasting sport league events among participating teams. A cooperative game theory framework is also adopted by Krumer and Moreno-Ternero (2023) to select participants in sport tournaments, with special focus on the allocation of slots to continental federations in the FIFA World Cup. In general, all these references suggest that fairness is not only an important aspect in sports settings, but also a complex issue to overcome.

In the next section, we present the TTP as well as some relevant concepts from cooperative game theory. In Section 3, we propose a methodology to find a tournament schedule by maximizing fairness. In Section 4 we study the compromise between the minimum distance and the maximum fairness solutions, based on the Pareto efficiency principle. In Section 5, we illustrate the application of the methodology and compare different solutions for the NL6 instance, one of the classic data instances of the TTP. Section 6 summarizes and concludes the article.

2. Preliminaries

In this section, we describe the traveling tournament problem (TTP) and present definitions of the relevant concepts from cooperative game theory.

2.1. Traveling tournament problem

In some sports leagues, the travel associated with visiting other teams' venues plays an important role due to the associated financial burden, the time spent on the road, and the effect of traveling fatigue in sport performance. In order to find schedules reducing the amount of travel, the TTP was introduced by Easton et al. (2001). Given an even number of teams n , the teams' locations, and values L and U , $L \leq U$, the TTP asks for a tournament schedule with $2(n-1)$ rounds where every pair of teams

plays each other twice, once at home and once away for each team, the number of consecutive games at home and consecutive games away is between L and U (inclusive), and the total distance traveled by the teams is minimized (assuming the teams do not get back home between consecutive away matches). The values of L and U are often set to 1 and 3, respectively.

A tournament in which every team faces each other team exactly twice, once at home and once away, is often referred to as a double round-robin tournament. Additionally, with n even and a total of $2(n-1)$ rounds, the solution of the TTP corresponds to a *compact* double round-robin tournament, i.e., the number of rounds is minimum and every team plays exactly once in every round. This is a common requirement in sports scheduling competitions (see e.g. Rasmussen and Trick (2008)). Besides, the TTP definition by Easton et al. (2001) introduces a *no-repeaters* condition.

We adopt the above specifications for the rest of this study. These summarize as follows.

- (C1) Every team faces each other team exactly twice, once at home and once away for each team.
- (C2) The number of rounds is minimum.
- (C3) No team can play more than three consecutive games at home or three consecutive games away.
- (C4) No two teams can play against each other in two consecutive rounds (no repeaters).

In order to determine a schedule satisfying conditions (C1)–(C4) while minimizing the total distance, we use a model based on the *formulation with $O(n^4)$ variables* described by Melo et al. (2009). With the set of teams N , the set of rounds $R = \{1, \dots, 2(n-1)\}$, and parameters d_{ij} stating the distance between venues of teams i and j for each combination $i, j \in N$, the model can be formulated as

$$\min \sum_{i \in N} \sum_{j \in N} \sum_{k \in R} d_{ij} x_{i,j,k} + \sum_{i \in N} \sum_{j \in N} \sum_{k \in R} d_{ji} x_{i,j,k} \quad (1)$$

$$\text{s.t.} \quad \sum_{j \in N, j \neq i} x_{i,j,k} = 1 \quad \forall t \in N, i \in N : i \neq t, \quad (2)$$

$$\sum_{i \in N} \sum_{j \in N} x_{i,j,k} = 1 \quad \forall t \in N, k \in R : k > 1, \quad (3)$$

$$\sum_{j \in N} x_{t,j,1} = 1 \quad \forall t \in N, \quad (4)$$

$$\sum_{j \in N} x_{t,i,j,k} = \sum_{j \in N} x_{t,j,i,k-1} \quad \forall t \in N, i \in N, k \in R : k > 1, \quad (5)$$

$$\sum_{j \in N} x_{i,j,t,k} = \sum_{i \in N, i \neq t} \sum_{j \in N, j \neq t} x_{i,j,t,k} \quad \forall t \in N, k \in R, \quad (6)$$

$$\sum_{j \in N} \sum_{i \in N, i \neq t} \sum_{l=0}^3 x_{t,i,j,k+l} \leq 3 \quad \forall t \in N, k \in R : k \leq |R| - 3, \quad (7)$$

$$\sum_{j \in N, j \neq t} \sum_{i \in N} \sum_{l=0}^3 x_{j,i,t,k+l} \leq 3 \quad \forall t \in N, k \in R : k \leq |R| - 3, \quad (8)$$

$$\sum_{j \in N} \sum_{l=0}^1 (x_{t,i,j,k+l} + x_{i,j,t,k+l}) \leq 1 \quad \forall t \in N, i \in N, k \in R : i \neq t, k < |R|, \quad (9)$$

$$x_{t,i,j,k} \in \{0, 1\} \quad \forall t \in N, i \in N, j \in N, k \in R. \quad (10)$$

The binary variable $x_{t,i,j,k}$ equals 1 if team $t \in N$ travels from venue of

team $i \in N$ to venue of team $j \in N$ in round $k \in R$. Hence, the objective function (1) minimizes the total distance traveled by all teams over all rounds (including the return home after the last game). Constraints (2), (3), (4), (5), and (6) ensure satisfaction of conditions (C1) and (C2) by prescribing each team to play at home against each other team once, play one game each round, start at home before the first round, travel in each round from the venue of the previous round's game, and meet another team for a game when staying home, respectively. Condition (C3) is enforced by constraints (7) and (8). For each team and for each sequence of four consecutive rounds, constraints (7) restrict the number of games away to be at most three. Constraints (8) work analogously for the home games. Constraints (9) make sure that no pair of teams plays against each other in two consecutive rounds, i.e., condition (C4). Lastly, (10) states the binary nature of the variables.

After solving model (1)–(10), the optimal objective value indicates the total distance traveled by the teams. The associated optimal schedule can be derived from the values of $x_{t,i,j,k}$.

2.2. Cooperative game theory

Let N denote the set of all players and $\mathcal{S}$ the set of all subsets of N . A transferable utility game (TU game) is a pair (N, c) where $c: \mathcal{S} \rightarrow \mathbb{R}$ is the characteristic function assigning to each coalition $S \in \mathcal{S}$ the optimal cost achievable by cooperation of players within this coalition. Analogously, the characteristic function can be defined as worth instead of cost, but we focus here on the cost version as the objective function (1) minimizes distances.

TU games allow for side payments, and thus for redistribution of the incurred costs among cooperating players. One of the central questions in cooperative game theory revolves around a fair distribution of the costs. Assuming that all players from N cooperate and incur a total cost $c(N)$, every player p should get allocated a cost q_p , such that $\sum_{p \in N} q_p = c(N)$. It is, in general, not trivial to decide how much of the total cost each player should bear, as in practice there is no consensus on what fairness actually means. In the same vein, the game-theoretic literature recognizes multiple cost allocation methods based on different views of fairness to determine such q_p values. Here, we describe three of the most commonly used: the egalitarian method, the Shapley value, and the nucleolus.

The egalitarian method allocates to every player the same share of $c(N)$, i.e., player p gets allocated a cost q_p defined as

$$q_p = \frac{c(N)}{|N|} \quad \forall p \in S, \quad (11)$$

where $|N|$ denotes the cardinality of N , i.e., the total number of players.

The Shapley value is another commonly used cost allocation method. [Algaba et al. \(2019\)](#) show its importance across many different applications. To achieve a fair distribution, Shapley ([Shapley et al., 1953](#)) derives the share of $c(N)$ assigned to player $p \in N$ from this player's marginal contributions to any possible coalition as

$$q_p = \sum_{S \subseteq N: p \in S} \frac{(|S| - 1)! (|N| - |S|)!}{|N|!} \cdot (c(S) - c(S \setminus \{p\})). \quad (12)$$

The nucleolus, as defined by [Schmeidler \(1969\)](#), takes a different approach to determine a fair distribution of $c(N)$. In this case, the shares are derived from excesses of the coalitions in order to minimize players' incentives to leave N and form another coalition. For $N = \{p_1, \dots, p_{|N|}\}$ and a cost allocation $q = (q_{p_1}, \dots, q_{p_{|N|}})$, the excess of coalition $S \subseteq N$ at q is defined as $e(q, S) = c(S) - \sum_{p \in S} q_p$. Furthermore, with $S_1, \dots, S_m$ denoting all subsets of N , $e(q) = (e(q, S_1), \dots, e(q, S_m))$ denotes the excess vector at q and $\theta(e(q))$ is a vector resulting from arranging the components of $e(q)$ in non-decreasing order. The nucleolus is then defined as the cost allocation q which lexicographically maximizes $\theta(e(q))$.

We remark here that there are many other cost allocation methods, for which we refer the reader to relevant literature on TU games. For

example, some methods attempt to find a compromise between two different methods, as studied in [Bergantiños and Moreno-Ternero \(2021\)](#) to share the revenues from broadcasting sports leagues. Other methods come from *ad hoc* definitions of fairness in specific context of applications, as surveyed by [Guajardo and Rönnqvist \(2016\)](#) to share the costs in collaborative logistics.

3. Fair scheduling in the TTP

The TTP looks for a schedule guaranteeing minimum total distance traveled by the teams. This does not imply minimum distance traveled by each particular team. In fact, some teams might need to travel more if it keeps the collective distance low. This observation suggests a question: considering the collective as well as all individual travels, what would a fair schedule look like? In the following, we determine fair schedules by, first, finding an *ideal* distance distribution and, then, determining the *nearest* schedule.

To find the ideal distance distribution, we employ the cost allocations introduced in the previous section. As most of the cost allocations require the characteristic function, in [Subsection 3.1](#), we define models to compute distances associated with each set of teams. With determined ideal distance distribution, we proceed by finding the nearest solution, a schedule in which the distances traveled by the teams resemble the distribution the most. [Subsections 3.2 and 3.3](#) focus on measuring of the resemblance and finding the nearest schedule, respectively.

3.1. Characteristic function

With N standing for the set of all teams in a tournament, we can define a TU game (N, c) . The characteristic function c assigns to each coalition $S \subseteq N$ the optimal traveling distance a tournament of teams within S could achieve. The computation of $c(S)$ varies depending on the properties of S .

First, we look at coalitions of two or less teams. For any $S \subseteq N$ such that $|S| = 0$ or $|S| = 1$, there are no games to be played. Hence, $c(S) = 0$. For any $S \subseteq N$ such that $|S| = 2$, exactly two games need to be played, where the two teams in S face each other at each team's venue. Thus, $c(S)$ equals the sum of the distances of two round-trips between these two venues (note for this particular case we must violate condition (C4)).

For any $S \subseteq N$ such that $|S|$ is even and $|S| > 2$, the value of $c(S)$ can be determined as the optimal objective value of model (1)–(10) with N replaced by S . For $S \subseteq N$ such that $|S|$ is odd and $|S| > 2$, the computation of $c(S)$ is not as straightforward. With an odd number of teams, in each round there has to be at least one team unmatched with an opponent. We refer to this situation as a team having a rest or having a bye as described for example by [Bao and Trick \(2010\)](#) in the case of the relaxed traveling tournament problem. With teams having byes, the minimum number of rounds no longer equals $2(|S| - 1)$ and model (1)–(10) cannot be applied. For instance, the combination of constraints (4) and (6) explicitly requires each team to play one game each round. The model thus needs to be modified. We assume that, when having a bye, the teams stay on the road, i.e., the teams do not travel home during a round with no game. We still require the model to follow conditions (C1)–(C4), but to avoid long periods without a game we introduce an additional condition as follows.

(C5) No team can have two or more consecutive byes.

To comply with condition (C2), the number of rounds needs to be minimum. With n teams in a tournament, each team plays $2(n - 1)$ games. This means a total of $n(n - 1)$ games in the tournament. With n even, a maximum number of games in any round equals $\frac{n}{2}$. With n odd, a maximum number of games in any round equals $\frac{n-1}{2}$. Therefore, the least number of rounds equals $2(n - 1)$ for n even and $2n$ for n odd. Hence, with an odd number of teams, each team needs to have two byes.

Using a general formulation for any $S \subseteq N$ such that $|S| > 1$, a model computing $c(S)$ while allowing for byes can be stated as follows.

$$c(S) = \min \sum_{t \in S} \sum_{i \in S} \sum_{j \in S} \sum_{k \in R} d_{i,j} x_{t,i,j,k} + \sum_{t \in S} \sum_{i \in S} \sum_{j \in S} d_{j,t} x_{t,i,j,|R|} \quad (13)$$

$$\text{s.t.} \quad \sum_{j \in S, j \neq t} \sum_{k \in R} x_{t,i,j,k} = 1 \quad \forall t \in S, i \in S : i \neq t, \quad (14)$$

$$\sum_{i \in S} \sum_{j \in S} x_{t,i,j,k} = 1 \quad \forall t \in S, k \in R : k > 1, \quad (15)$$

$$\sum_{j \in S} x_{t,i,j,1} = 1 \quad \forall t \in S, \quad (16)$$

$$\sum_{j \in S} x_{t,i,j,k} = \sum_{j \in S} x_{t,i,j,k-1} \quad \forall t \in S, i \in S, k \in R : k > 1, \quad (17)$$

$$\sum_{j \in S} x_{t,i,j,k} \geq \sum_{i \in S, i \neq t} \sum_{j \in S, j \neq t} x_{i,j,t,k} \quad \forall t \in S, k \in R, \quad (18)$$

$$\sum_{j \in S} \sum_{i \in S, i \neq t} \sum_{m \in R, k \leq m < k+l} x_{t,i,j,m} \leq 3 + \sum_{j \in S} \sum_{i \in S, i \neq t} \sum_{m \in R, k \leq m < k+l} M x_{j,i,t,m} \quad \forall t \in S, k \in R, l \in R : k \leq |R| - 3, 4 \leq l \leq |R| - k + 1, \quad (19)$$

$$\sum_{j \in S, j \neq t} \sum_{i \in S, i \neq t} \sum_{m \in R, k \leq m < k+l} x_{j,i,t,m} \leq 3 + \sum_{j \in S} \sum_{i \in S, i \neq t} \sum_{m \in R, k \leq m < k+l} M x_{j,i,t,m} \quad \forall t \in S, k \in R, l \in R : k \leq |R| - 3, 4 \leq l \leq |R| - k + 1, \quad (20)$$

$$\sum_{j \in S, j \neq t} \sum_{l=0}^1 x_{t,i,j,k+l} + \sum_{j \in S, j \neq t} \sum_{l=0}^1 x_{t,i,j,k+l} \leq 1 \quad \forall t \in S, i \in S, k \in R : i \neq t, k < |R|, |R| > 2, \quad (21)$$

$$\sum_{l=0}^1 x_{t,i,j,k+l} + \sum_{j \in S, j \neq t} \sum_{l=0}^1 x_{t,i,j,k+l} - \sum_{j \in S, j \neq t} \sum_{h \in S, h \neq t} \sum_{l=0}^1 x_{j,h,t,k+l} \leq 1 \quad \forall t \in S, i \in S, k \in R : k < |R|, \quad (22)$$

$$x_{t,i,j,k} \in \{0, 1\} \quad \forall t \in S, i \in S, j \in S, k \in R, \quad (23)$$

where $R = \{1, \dots, 2(n-1)\}$ for $|S|$ even or $R = \{1, \dots, 2n\}$ for $|S|$ odd and M is a sufficiently large number, for example $M = |R|$.

The binary variables $x_{t,i,j,k}$ in model (13)–(23) work in the same way as in model (1)–(10). If team $t \in N$ travels from a venue of team $i \in N$ to play a game at a venue of team $j \in N$ in round $k \in R$, $x_{t,i,j,k}$ equals 1. Additionally, if team $t \in N$ is to have a rest after being at a venue of team $i \in N$, $x_{t,i,i,k}$ equals 1. The objective function (13) and constraints (14)–(17), (23) are equivalent to the objective function (1) and constraints (2)–(5), (10), respectively, with the only difference being the replacement of set N by S . Constraints (18) state that a team may travel to another team's venue for a game on a given round only if the corresponding host team plays a home game in such round. Constraints (19) and (20) guarantee compliance with condition (C3). If any sequence of at least four consecutive rounds includes no game at home, constraints (19) require the number of games away in this sequence to be at most three. Constraints (20) work analogously to restrict the number of consecutive games at home. Constraints (21) prevent teams from playing against the same opponent in two consecutive rounds, i.e., condition (C4). Additionally, condition $|R| > 2$ in constraints (21) avoids infeasibility of the model for tournaments of less than 3 teams. Lastly, for each team,

constraints (22) enforce condition (C5) by restricting the number of rests in any two consecutive rounds to be at most one.

Model (13)–(23) can be solved for any coalition $S \subseteq N$ regardless of its cardinality or parity of this cardinality. With fully specified characteristic function c , it is then easy to determine cost allocations such as the egalitarian method, the Shapley value or the nucleolus. Since these allocations follow certain concepts of fairness, they may serve as the ideal distance distribution.

3.2. Measuring fairness

With the knowledge of the ideal distance distribution, we would like to find a schedule that enables all teams to travel these ideal distances. However, since the space of feasible solutions to the scheduling problem is discrete, it might become infeasible to construct such a schedule. Thus, we focus on finding a schedule which resembles the ideal distance distribution the most. Since we consider resemblance regarding only the distances traveled by the teams, we can represent each schedule by a vector of the distances. Hence, when we compare the ideal distance distribution with a schedule or a schedule with another schedule, we essentially compare two vectors of length $|N|$, where the components of the vector represent the distances traveled by each team. To perform the comparison, we focus here on the RD_a metric proposed by [Perea and Puerto \(2019\)](#), who develop a heuristic procedure to compute the nucleolus and use this metric to compare the actual nucleolus with the heuristic solution. For a given allocation $q = (q_1, \dots, q_n)$ and the nucleolus $\bar{q} = (\bar{q}_1, \dots, \bar{q}_n)$, the comparison metric is defined as follows:

$$RD_a = \frac{\sqrt{\sum_{j=1}^n (\bar{q}_j - q_j)^2}}{\sqrt{\sum_{j=1}^n \bar{q}_j^2}} \quad (24)$$

Note that in computing the RD_a metric, the sign of the difference in $\sum_{j=1}^n (\bar{q}_j - q_j)^2$ is not relevant. This might be questionable in our context, where the case that most matters is when the prescribed distances are higher than the ideal ones (that is, when the difference $(\bar{q}_j - q_j)$ is negative). For illustration, let us assume a four-team tournament with ideal distance distribution (150, 100, 200, 80) and only two feasible schedules prescribing distances (300, 80, 210, 75) and (300, 95, 210, 75). In both solutions, the first, third, and fourth team face the same distances. On the other hand, the second team could benefit from the first solution even though the distance prescribed by the second solution is closer to the ideal one. Nevertheless, due to the closer distance in the second solution, this would be the preferred schedule from the perspective of the RD_a metric. Hence, pursuing the RD_a metric could make a particular team travel a longer distance even if no other team would benefit from it. This can be avoided by focusing only on the negative differences. Therefore, we propose a modification of the RD_a metric accounting only for the negative differences, which we denote by RD_a^- and define as follows:

$$RD_a^- = \frac{\sqrt{\sum_{j=1}^n \tilde{q}_j - q_j < 0 (\bar{q}_j - q_j)^2}}{\sqrt{\sum_{j=1}^n \bar{q}_j^2}} \quad (25)$$

3.3. Maximizing fairness

With a set of teams $N = \{t_1, \dots, t_{|N|}\}$, and provided that the characteristic function values have already been computed, it is not hard to obtain the ideal distance distribution $\tilde{x} = (\tilde{x}_{t_1}, \dots, \tilde{x}_{t_{|N|}})$ for the traditional allocation methods. Then, with a chosen metric of fairness, to find the nearest solution (schedule maximizing the fairness), we can focus on finding a schedule that minimizes this metric. In particular, for the RD_a metric, the nearest solution can be determined by a model with objective function

$$\min \frac{\sqrt{\sum_{t \in N} \left(\tilde{x}_t - \sum_{i \in S} \sum_{j \in S} \sum_{k \in R} d_{ij} x_{t,ij,k} - \sum_{i \in S} \sum_{j \in S} d_{j,t} x_{t,ij,|R|} \right)^2}}{\sqrt{\sum_{t \in N} \tilde{x}_t^2}}, \quad (26)$$

while satisfying constraints (14)–(23) in which all occurrences of set S are replaced by N . We refer to this model as the RD_a -minimizing TTP.

Analogously, for the RD_a^- metric, we define the RD_a^- -minimizing TTP replacing the objective function by the following:

$$\min \frac{\sqrt{\sum_{t \in N: D_t < 0} D_t^2}}{\sqrt{\sum_{t \in N} \tilde{x}_t^2}}, \quad (27)$$

where

$$D_t = \tilde{x}_t - \sum_{i \in S} \sum_{j \in S} \sum_{k \in R} d_{ij} x_{t,ij,k} - \sum_{i \in S} \sum_{j \in S} d_{j,t} x_{t,ij,|R|} \quad \forall t \in N. \quad (28)$$

In both objective functions above the denominators do not depend on the variables of the models and, thus, remain constant. Therefore, and because the square root function is a strictly increasing function, the objective functions (26), and (27) may be replaced by

$$\min \sum_{t \in N} \left(\tilde{x}_t - \sum_{i \in S} \sum_{j \in S} \sum_{k \in R} d_{ij} x_{t,ij,k} - \sum_{i \in S} \sum_{j \in S} d_{j,t} x_{t,ij,|R|} \right)^2, \quad (29)$$

and

$$\min \sum_{t \in N: D_t < 0} D_t^2, \quad (30)$$

respectively, with no change in the optimal solutions. Despite the simplification, these objective functions remain nonlinear. With the focus of our paper not being algorithmic, we have limited ourselves to the use of optimization solvers to deal with these problems. In particular, the RD_a -minimizing TTP and the RD_a^- -minimizing TTP both lead to an integer quadratic programming problem solvable by solvers such as Gurobi or CPLEX.

4. Pareto efficiency

As studied in the previous section, the TTP selects a schedule associated with the shortest total distance. Maximizing fairness, on the other hand, results in a schedule minimizing a given metric. Although the solutions might coincide, it is not always the case. Since both criteria are often desirable, it is not trivial to decide which schedule is the best. To aid this decision, in this section we use a method from multi-objective optimization called the Pareto front.

The RD_a -minimizing TTP, RD_a^- -minimizing TTP, and model (13)–(23) for $S = N$ contain the same set of constraints and differ only in their objective functions. Hence, their feasible regions are identical and, for a given ideal distance distribution, values of all the objective functions can be evaluated at each feasible solution.

As described for example by Hwang and Masud (1979), for a problem with two objective functions, a solution is considered Pareto efficient if none of the objective functions can be improved without worsening the other one. Pareto front is then the set of all Pareto efficient solutions.

For example, let us assume selection of the RD_a^- metric in a problem with eight feasible schedules depicted in Fig. 1. Only the solutions denoted by a full circle belong to the Pareto front. For each of the other solutions, there exists another solution improving at least one of the objectives without worsening the other one. Therefore, it is reasonable to choose a schedule belonging to the Pareto front. The left-most solution on the Pareto front is the schedule minimizing RD_a^- , while the right-most solution corresponds to the minimization of the total distance. The two

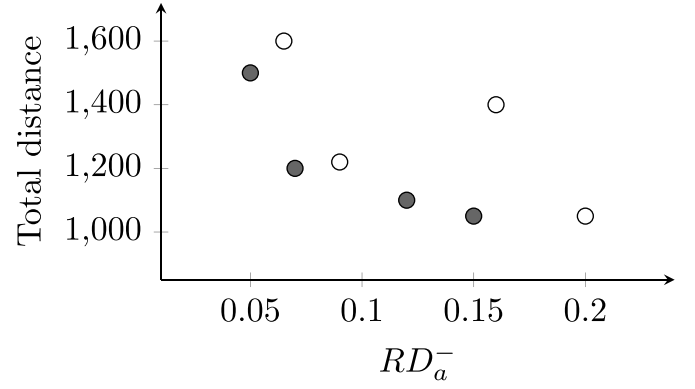


Fig. 1. Example of a Pareto front.

remaining solutions offer some compromise between the two criteria.

5. Numerical example

In this section, to illustrate the proposed methodology, we attempt to find a fair schedule in the NL6 instance by Easton et al. (2001). All our models were implemented in AMPL and solved with the Gurobi solver.

The NL6 instance contains six teams from the North American baseball league known as the National League. Fig. 2 describes the teams and location of their venues.

In the rest of this section, we refer to the teams only by their handles.

In Fig. 2b, one can notice that teams NYM and PHI are based fairly close to each other. Teams FLA and ATL are on the other hand quite distanced from all other teams. The notion of fairness might vary across different leagues and tournaments. For instance, one could see team FLA in an unfortunate position given its location and seek balance in traveled distances by minimizing this team's distance on the expense of other teams. This viewpoint would correspond with a choice of the egalitarian method or a similar allocation. Nonetheless, one could also look at centrally located teams like NYM and PHI and claim that they should not be punished for other teams such as team FLA, i.e., the teams should travel in line with marginal distances they bring to the tournament. In diverging from the egalitarian solution, concepts such as the Shapley value, the nucleolus or other allocation methods become important.

We proceed by solving the TTP model (1)–(10). The optimal schedule prescribes distances shown in the first part of Table 1. Afterwards, model (13)–(23) is solved for each $S \subseteq N$ in order to evaluate the characteristic function which is then used to compute the egalitarian method, Shapley value and nucleolus. These are outlined in Table 1. The egalitarian method prescribes each team the same distance of 3, 986. The Shapley value and the nucleolus, as expected from the geographical distribution of the teams' venues, prescribe the lowest distances to teams NYM and PHI with no big difference while the distance determined for team FLA is more than twice as large.

To find schedules with distances similar to the three computed allocations, we can now solve the RD_a -minimizing TTP and the RD_a^- -minimizing TTP. The results are listed in the last part of Table 1. Although the schedules do not match the allocations exactly, they tend to be more similar to the respective allocations than the solution of the original TTP.

In Subsection 3.2, we have discussed the main advantage of using the RD_a^- metric over the RD_a metric. In this example, for the egalitarian method and the nucleolus, the models actually result in the same solutions for both of the metrics. This is not the case with the Shapley value. The distances determined by the models differ for all teams except for PHI. In case of ATL, NYM, MON, and PIT the distances are still fairly similar. The issue occurs in the case of team FLA. While the RD_a^- metric leads to a distance of 5, 596, the RD_a metric determines a solution with a distance of 6, 407. Although this is much closer to the value of 6, 336 determined by the Shapley value, team FLA would certainly prefer the

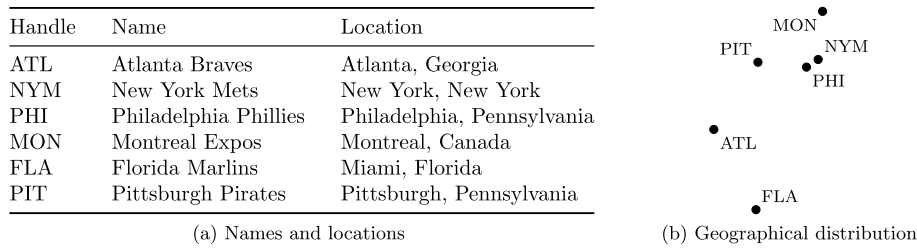


Fig. 2. Description of teams from NL6 instance.

Table 1

Distance allocations across different methods and metrics.

	ATL	NYM	PHI	MON	FLA	PIT	Total distance
TTP	4, 414	3, 328	3, 724	3, 996	5, 135	3, 319	23, 916
Egalitarian method	3, 986	3, 986	3, 986	3, 986	3, 986	3, 986	23, 916
Shapley value	3, 781.1	3, 141.3	3, 008.3	4, 488.8	6, 336	3, 161.5	23, 916
Nucleolus	4, 176.8	2, 938.5	2, 893.5	4, 117.5	6, 715.3	3, 074.5	23, 916
RD_a^- -min. TTP (egal. m.)	4, 147	3, 857	4, 174	4, 090	4, 953	4, 058	25, 279
RD_a^- -min. TTP (egal. m.)	4, 147	3, 857	4, 174	4, 090	4, 953	4, 058	25, 279
RD_a^- -min. TTP (Shapley v.)	4, 263	3, 472	3, 372	4, 424	6, 407	3, 597	25, 535
RD_a^- -min. TTP (Shapley v.)	4, 147	3, 365	3, 372	4, 537	5, 596	3, 604	24, 621
RD_a^- -min. TTP (nucleolus)	4, 551	3, 509	3, 200	3, 994	6, 455	3, 218	24, 927
RD_a^- -min. TTP (nucleolus)	4, 551	3, 509	3, 200	3, 994	6, 455	3, 218	24, 927

first option. It hence seems like an unnecessary increase in the distance for team FLA which is also reflected in the total distance. This supports the discussion in Subsection 3.2 and, therefore, for a detailed analysis of the results, we proceed only with the models based on the RD_a^- metric.

Fig. 3 reports the optimal schedules obtained by solving the TTP and the RD_a^- -minimizing TTP for all three allocations.

(character @ preceding a team handle indicates an away game). Although the schedules, as well as the distances, differ significantly, one can notice some similar patterns. For instance, the schedules of team ATL in Fig. 3b and c imply the same distance despite coinciding only in rounds 1, 9 and 10. This demonstrates the flexibility in scheduling and shows that accommodating one team does not necessarily have a negative effect on all other teams.

To better understand how the different schedules result in different distances, in Fig. 4, we display the actual travel of team FLA whose distances vary the most across the optimal schedules.

This makes the travel of team FLA ideal for illustrative purposes. For the interested reader, figures depicting the travel of all other teams are provided in Appendix A. The TTP results with FLA covering a distance of 5, 135. This is achieved by two trips, first visiting teams ATL and PHI and second visiting teams NYM, MON and PIT. Although it might look counterintuitive that teams PHI and NYM are visited in separate trips (as shown in Fig. 4a, these are very close to each other), note that when FLA visits PHI, the NYM team is on a road trip so it could not host a home match. In contrast, with the RD_a^- -minimizing TTP and the egalitarian method, the opportunity for FLA to visit PHI in NYM in consecutive matches is used, and the distance drops to 4, 953. This is in fact the shortest possible option for team FLA. Hence, even though the distance is still much larger than the one prescribed by the egalitarian method, it has got as close as it could. The Shapley value and the nucleolus, on the other hand, prescribe team FLA to cover longer distances than 5, 135. The RD_a^- -minimizing TTP's with respect to these allocations indeed increase the distance. Fig. 4c and d shows the associated travel. One could argue that the travel seems very inefficient. However, since we are using the RD_a^- metric, this inefficiency and increase in distance is a result of other teams' decreasing distances to achieve a more fair solution. This example illustrates well the utility of introducing the RD_a^- metric instead of the more basic RD_a metric.

In Table 1, we can see that, for all allocations, the RD_a^- -minimizing TTP is associated with an increase in the total distance. Therefore, as a

Round											
Team	1	2	3	4	5	6	7	8	9	10	Distance
ATL	FLA	NYM	PIT	@PHI	@MON	@PIT	PHI	MON	@NYM	@FLA	4,414
NYM	@PIT	@ATL	@FLA	MON	FLA	@PHI	@MON	PIT	ATL	PHI	3,328
PHI	@MON	FLA	MON	ATL	@PIT	NYM	@ATL	@FLA	PIT	@NYM	3,724
MON	PHI	@PIT	@PHI	@NYM	ATL	FLA	NYM	@ATL	@FLA	PIT	3,996
FLA	@ATL	@PHI	NYM	PIT	@NYM	@MON	@PIT	PHI	MON	ATL	5,135
PIT	NYM	MON	@ATL	@FLA	PHI	ATL	FLA	@NYM	@PHI	@MON	3,319
Total distance											23,916

(a) Optimal solution of the TTP

Round											
Team	1	2	3	4	5	6	7	8	9	10	Distance
ATL	FLA	MON	@PHI	@NYM	@MON	PIT	PHI	NYM	@PIT	@FLA	4,147
NYM	PIT	PHI	@PIT	ATL	@PHI	MON	FLA	@ATL	@FLA	@MON	3,857
PHI	MON	@NYM	ATL	@FLA	NYM	FLA	@ATL	@PIT	@MON	PIT	4,174
MON	PHI	@ATL	@FLA	PIT	ATL	@NYM	@PIT	FLA	PHI	NYM	4,090
FLA	@ATL	@PIT	MON	PHI	PIT	@PHI	@NYM	@MON	NYM	ATL	4,953
PIT	@NYM	FLA	NYM	@MON	@FLA	@ATL	MON	PHI	ATL	@PHI	4,058
Total distance											25,279

(b) Optimal solution of the RD_a^- -minimizing TTP (egalitarian method)

Round											
Team	1	2	3	4	5	6	7	8	9	10	Distance
ATL	FLA	NYM	PIT	@MON	@NYM	@PHI	MON	PHI	@PIT	@FLA	4,147
NYM	@PIT	@ATL	@FLA	PHI	ATL	@MON	FLA	PIT	MON	@PHI	3,365
PHI	MON	@PIT	@MON	@NYM	FLA	ATL	PIT	@ATL	@FLA	NYM	3,372
MON	@PHI	FLA	PHI	ATL	@PIT	NYM	@ATL	@FLA	@NYM	PIT	4,537
FLA	@ATL	@MON	NYM	PIT	@PHI	@PIT	@NYM	MON	PHI	ATL	5,596
PIT	NYM	PHI	@ATL	@FLA	MON	FLA	@PHI	@NYM	ATL	@MON	3,604
Total distance											24,621

(c) Optimal solution of the RD_a^- -minimizing TTP (Shapley value)

Round											
Team	1	2	3	4	5	6	7	8	9	10	Distance
ATL	@FLA	PHI	MON	NYM	@PIT	@MON	@NYM	PIT	@PHI	FLA	4,551
NYM	@MON	FLA	PIT	@ATL	@FLA	@PIT	ATL	PHI	MON	@PHI	3,509
PHI	@PIT	@ATL	@FLA	PIT	MON	FLA	@MON	@NYM	ATL	NYM	3,200
MON	NYM	PIT	@ATL	@FLA	@PHI	ATL	PHI	FLA	@NYM	@PIT	3,994
FLA	ATL	@NYM	PHI	MON	NYM	@PHI	@PIT	@MON	PIT	@ATL	6,455
PIT	PHI	@MON	@NYM	@PHI	ATL	NYM	FLA	@ATL	@FLA	MON	3,218
Total distance											24,927

(d) Optimal solution of the RD_a^- -minimizing TTP (nucleolus)

Fig. 3. Optimal schedules.

last step, we can draw the Pareto fronts. These are displayed in Fig. 5 where each circle represents two schedules.

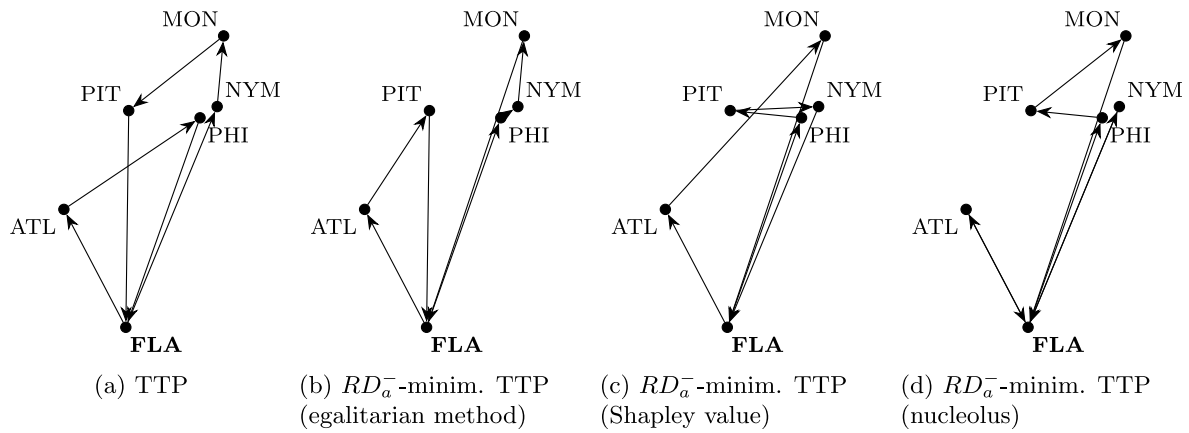


Fig. 4. Travel of team FLA.

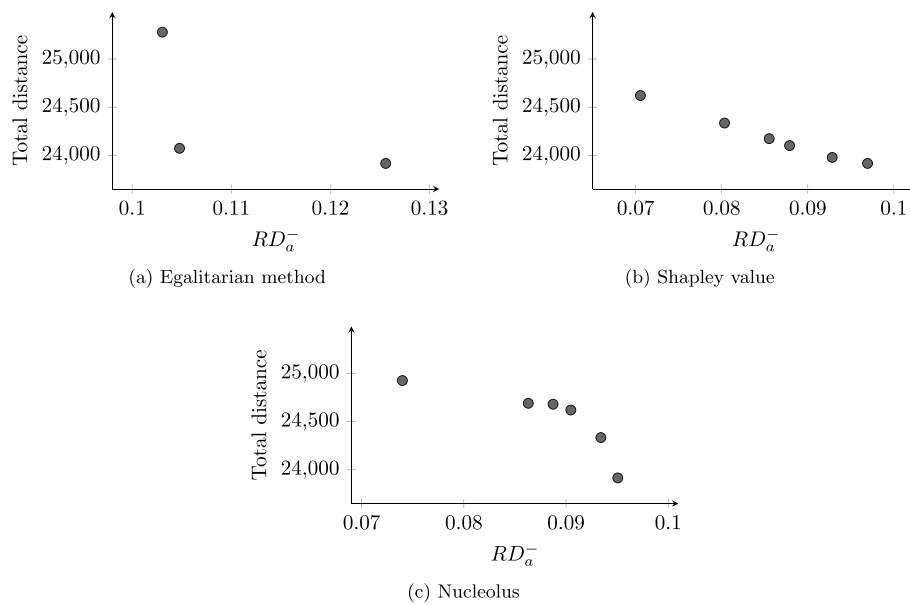


Fig. 5. Pareto fronts.

This is due to the fact that each feasible schedule can be reversed and run from the last round to the first round without violating any constraints while resulting in the same distances.

In the case of the egalitarian method, one can see in Fig. 5a that, although the upper-left circle minimizes the RD_a^- metric and the lower-right circle minimizes the total distance, the lower-left circle performs very well on both of the criteria. Therefore, a schedule corresponding with the lower-left circle seems to be a reasonable choice. In Fig. 5b and c, it is less straightforward as some solutions appear closer to each other. In general, the Pareto front offers some options to choose from and help understand the trade-offs. There is no general recipe for what is the right choice from the schedules on the Pareto front. While a practical approach would very much depend on the tournament itself as well as on available resources, and the inclination towards fairness instead of efficiency, the main conclusion of our analysis is that a schedule optimizing the overall distance traveled in total by all teams, in general, does not adhere to theoretical principles of fairness.

6. Conclusion

In this article, we have developed a cooperative game theory approach to produce fair tournament schedules in sports competitions, with particular focus on the celebrated TTP. To our knowledge, this is the

first research effort combining sports scheduling and cooperative game theory concepts. Given the importance of scheduling design in sports competitions and the importance of fairness in the outcomes and in the perception of the stakeholders around sport competitions, the study of such well-established concepts of fairness in well-established problems is relevant for the progress of the area.

To determine an ideal distribution of travel distances, we have used three different methods. As illustrated by our numerical results, the choice of a specific method is crucial for the final schedule. Since the different methods are based on different notions of fairness, this choice should reflect tournament planners' perception of what fairness in a given tournament means.

Next, to find the nearest schedule to a particular distance distribution, we have incorporated the use of some metrics to measure the resemblance. Overall, we have shown how to incorporate fairness to tournament scheduling as well as how to arrive at a set of solutions representing a compromise between fairness and minimum total distance. The resulting optimization problems are challenging, but this is precisely one of the main reasons why the TTP has stimulated large research efforts. Moreover, although our focus has been on the TTP, the framework for finding an ideal distance distribution and a nearest schedule is general enough to adjust to different tournament settings. For example, in single round-robin tournaments, the travel distances of the different teams may

differ from one to another schedule, thus selecting a fair schedule is relevant. In double round-robin tournaments where the teams get back home after away games (so the travel distances are constant), other objective functions become relevant, such as the minimization of penalties or maximization of attractive matches in suitable rounds. Our approach can also be extended to study such situations and, in general, to address other fairness concerns in sports scheduling. In this regard, we remark that the input to define the cooperative game is a result of the underlying scheduling problem, regardless of whether it is the TTP or another problem. In turn, the computation of this input is often hard in larger-size instances and the computation time increases exponentially with the number of teams, thus developing approximate solution methods remains of interest. On the applied side, it is particularly interesting to analyze how scheduling considering fairness could help teams in low-tier leagues, where resources are likely to be very limited and travel costs account for a large part of total expenses.

Finally, although transferable utility games have gained popularity to address cooperative problems in a variety of fields, there is also a non-transferable utility framework. Extending our work by this framework

may also be of interest, as traveling fatigue and recovery of players while teams are on the road intrinsically involves aspects of non-transferable utility.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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During the revision of this article, our former colleague Kurt Jörnsten unfortunately passed away. Although not directly linked to this paper, he was our co-author in other articles about cooperative game theory and shared many ideas with us. We remain deeply thankful to him for the entertaining discussions and for being a truly passionate and inspiring researcher.

Appendix A. Teams' travel for different schedules

For the different schedules, Figs. 6, 7, 8, 9, and 10 depict the actual travel of teams ATL, NYM, PHI, MON, and PIT, respectively. Similar conclusions could be drawn from these figures as in the case of team FLA in Section 5.

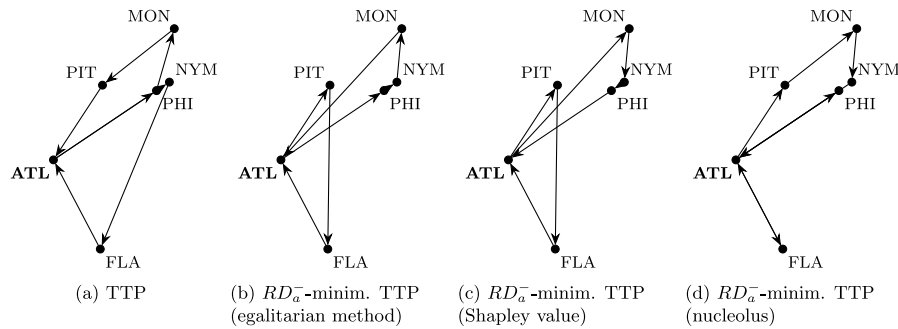


Fig. 6. Travel of team ATL.

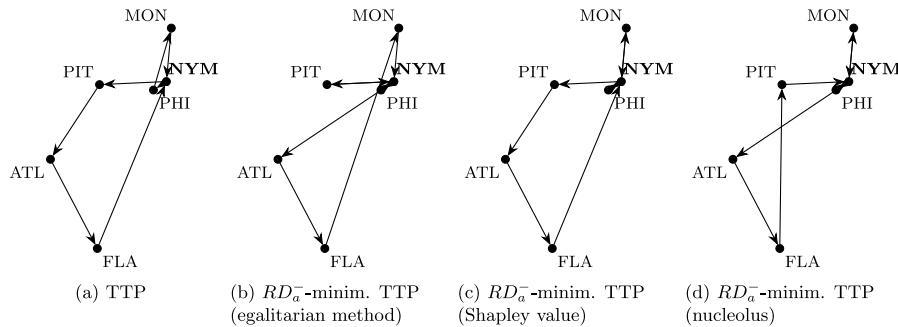


Fig. 7. Travel of team NYM.

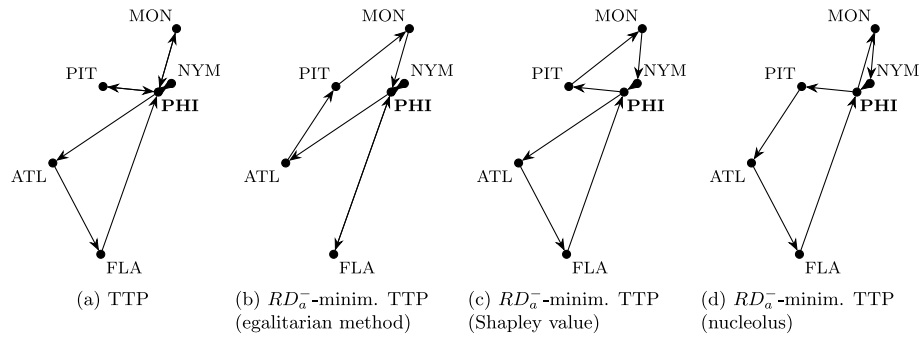


Fig. 8. Travel of team PHI.

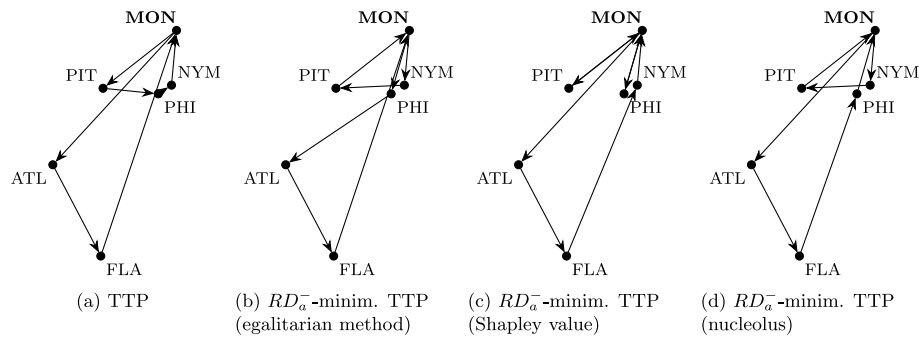


Fig. 9. Travel of team MON.

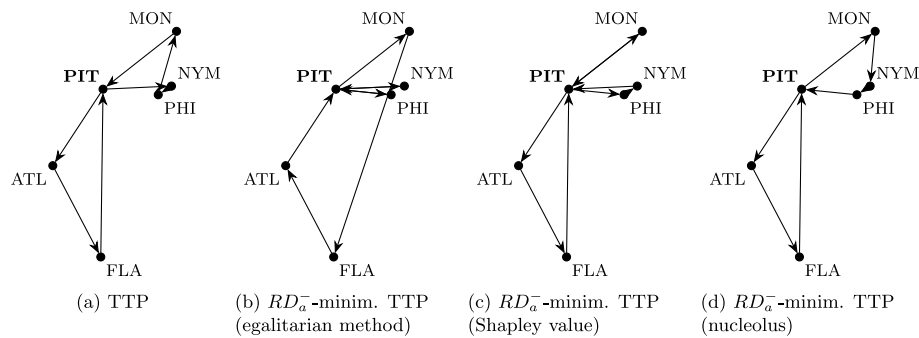


Fig. 10. Travel of team PIT.

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